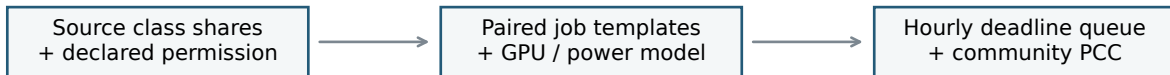


a

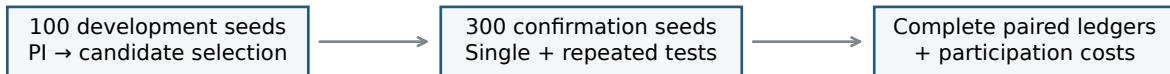
From workload composition to electrical response



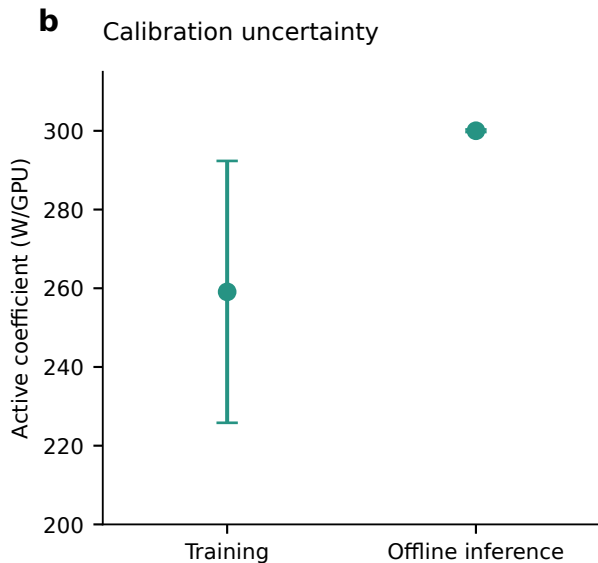
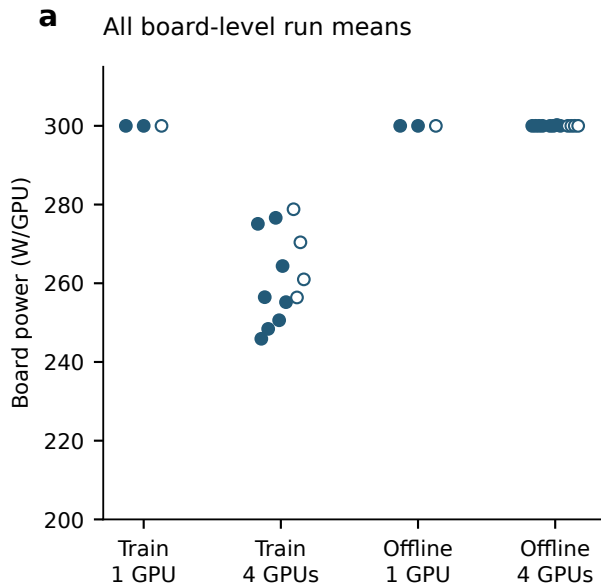
5%, 10%, 20%: source mix retained 40%: wider opt-in 60%: changed business mix

b

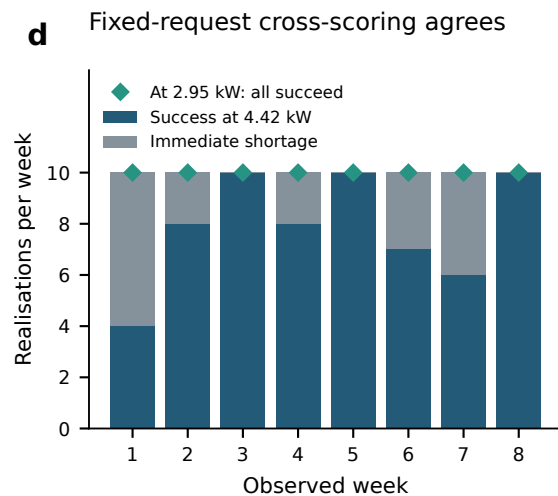
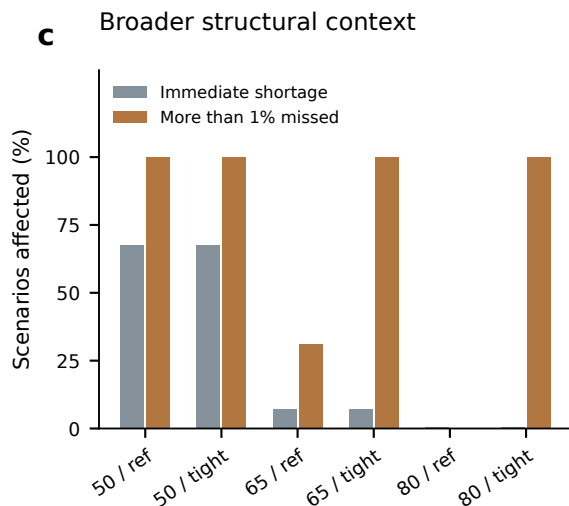
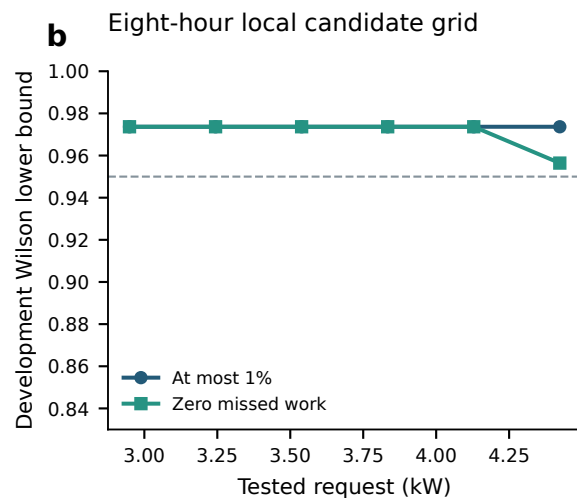
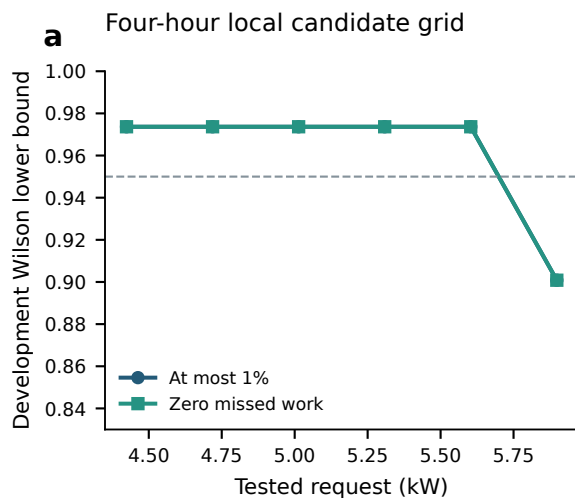
Evidence flow and independent qualification



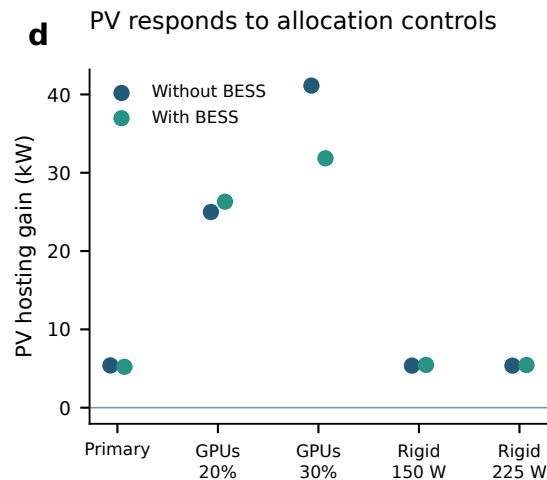
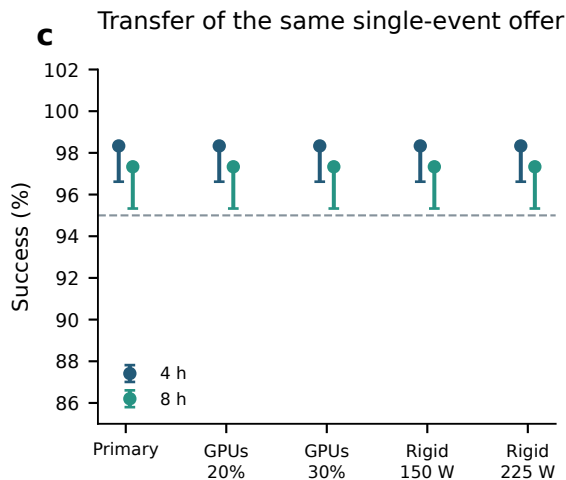
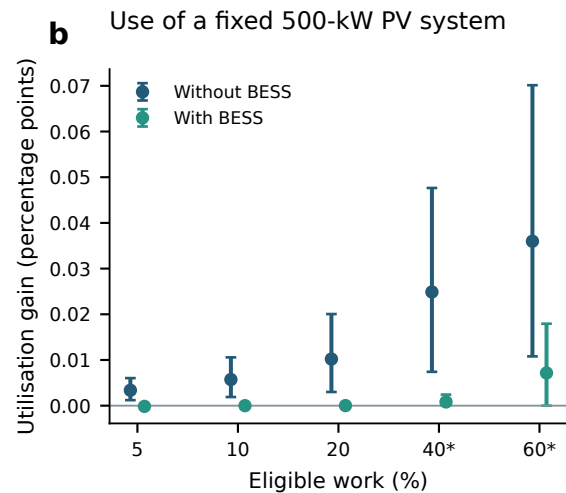
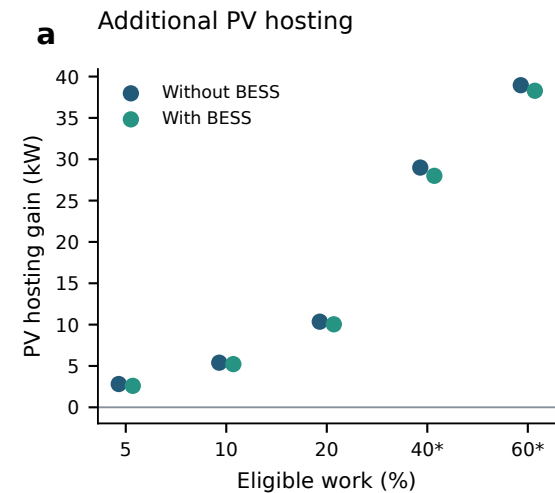
Separate full-information branch: PV hosting and utilisation, with / without BESS (100 seeds)



Filled points: fitting runs 1-2; open: held-out run 3. Boards within a run are not independent repeats.
Intervals: 95% Student t from two independent four-GPU run means. Overall held-out MAE: 3.80 W/GPU.



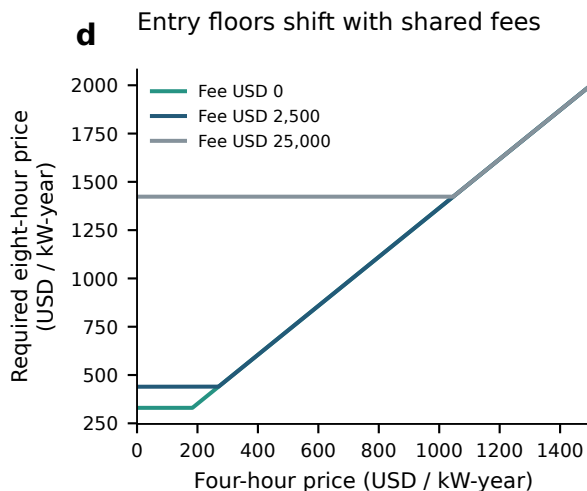
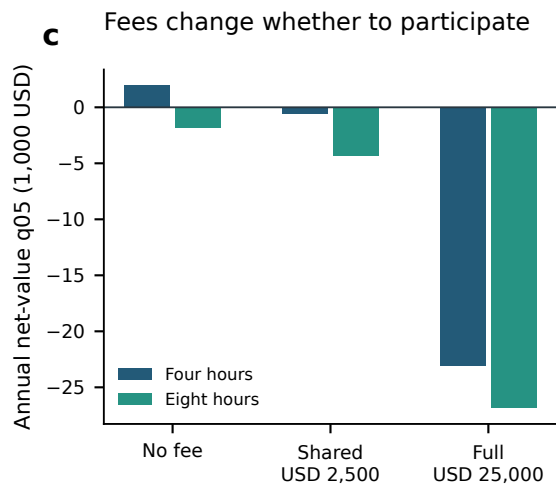
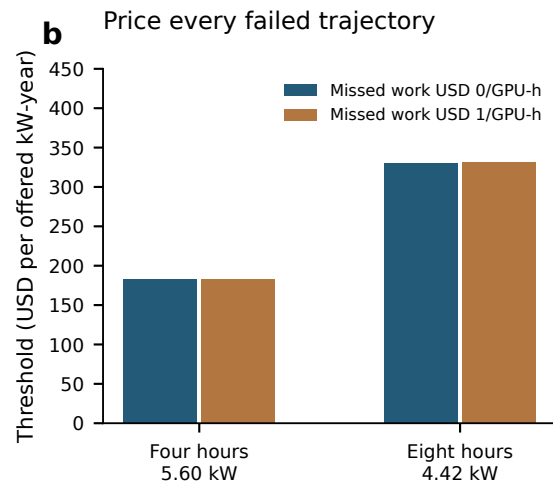
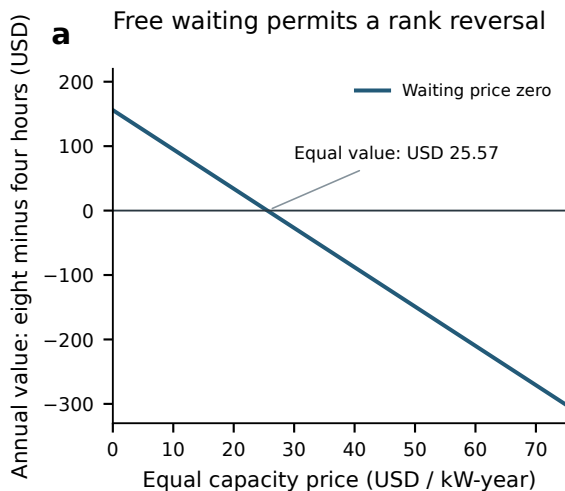
a,b: $n = 100$ independent development scenarios per candidate; finite grid. c: $n = 300$; failure families may overlap.
d: separate 986000-series count test; both service scores coincide at each request; no binomial confidence inference.
Matched last-call outcomes remain in Table 15; historical selection and score interpretation are in Supplementary Note 7.



a,b: five permission scenarios; *40% and 60% require additional business assumptions. PV: $n = 100$, zero missed work.

b: paired 95% bootstrap intervals cover sampling, not optimisation error; tiny BESS effects are unresolved.

c,d: fixed 10% work eligibility. c: $n = 300$; one-sided 95% Wilson bounds. Hosting gains are differences of minima.



Refined zero-loss products; same reference workload and annual accounting as Figure 6. All 300 trajectories retained.

a: positive values favour eight hours; b-d: waiting USD 0.005/GPU-h/h; c: equal capacity prices USD 250/kW-year.

Shared access fees do not pool reliability. These are stated-price accounting scenarios, not measured participation.

a Joint success: $\leq 1\%$ missed work

Utilisation (%) / deadline / GPU share (%)	Fresh last call	Four calls 16-h starts	Four calls 24-h starts
50 / 1x / 10	223	0	80
50 / ½x / 10	220	0	0
65 / 1x / 10	292	190	190
65 / ½x / 10	292	0	0
65 / ½x / 20	292	0	0
80 / 1x / 10	298	298	245
80 / ½x / 10	298	0	0
80 / ½x / 20	298	0	4

b Joint success: zero missed work

Utilisation (%) / deadline / GPU share (%)	Fresh last call	Four calls 16-h starts	Four calls 24-h starts
50 / 1x / 10	217	0	56
50 / ½x / 10	83	0	0
65 / 1x / 10	292	30	153
65 / ½x / 10	241	0	0
65 / ½x / 20	241	0	0
80 / 1x / 10	298	266	205
80 / ½x / 10	291	0	0
80 / ½x / 20	291	0	0

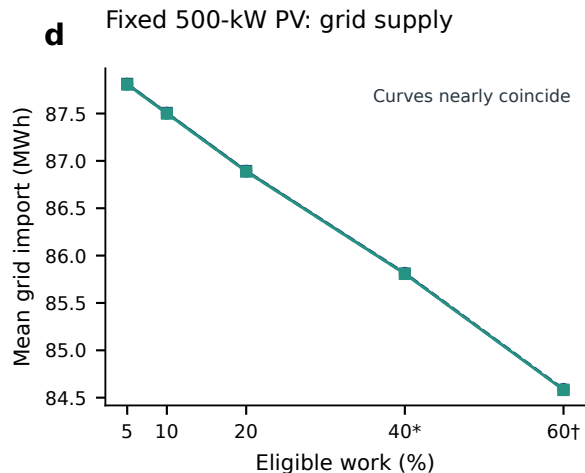
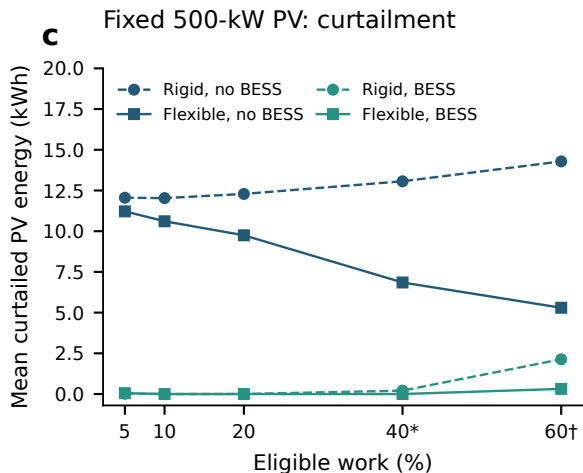
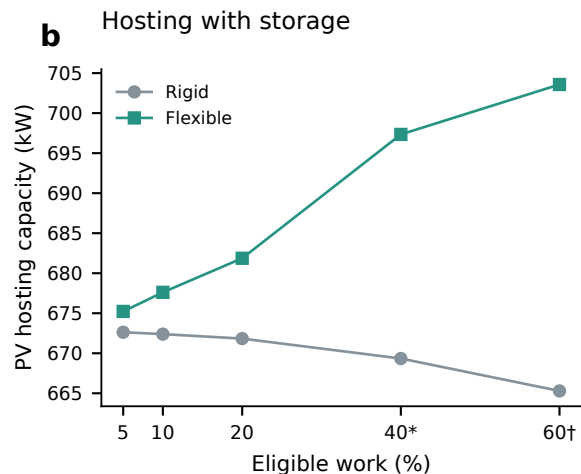
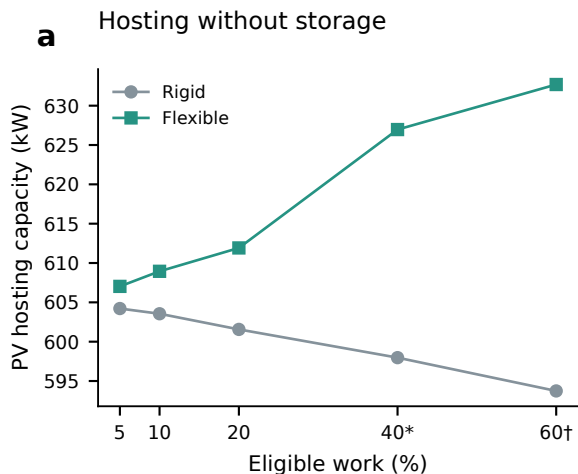
c Immediate supply shortage

Utilisation (%) / deadline / GPU share (%)	Fresh last call	Four calls 16-h starts	Four calls 24-h starts
50 / 1× / 10	77	203	155
50 / ½× / 10	77	203	155
65 / 1× / 10	8	22	22
65 / ½× / 10	8	22	22
65 / ½× / 20	8	22	22
80 / 1× / 10	2	2	3
80 / ½× / 10	2	2	3
80 / ½× / 20	2	2	3

d More than 1% missed work

Utilisation (%) / deadline / GPU share (%)	Fresh last call	Four calls 16-h starts	Four calls 24-h starts
50 / 1x / 10	0	300	162
50 / ½x / 10	3	300	300
65 / 1x / 10	0	93	96
65 / ½x / 10	0	300	300
65 / ½x / 20	0	300	300
80 / 1x / 10	0	0	52
80 / ½x / 10	0	300	300
80 / ½x / 20	0	300	296

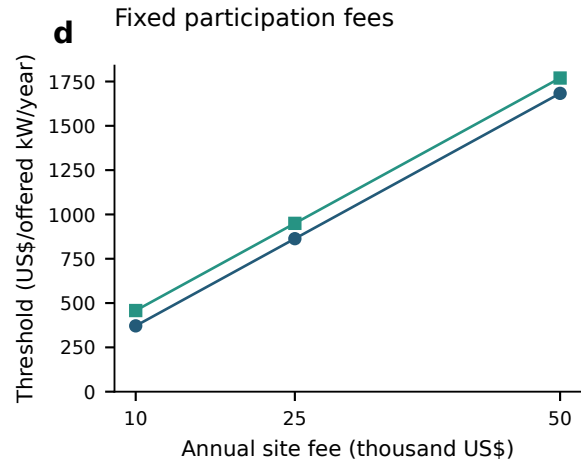
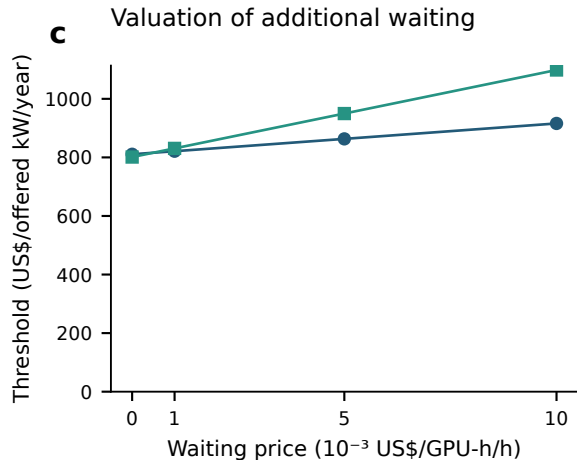
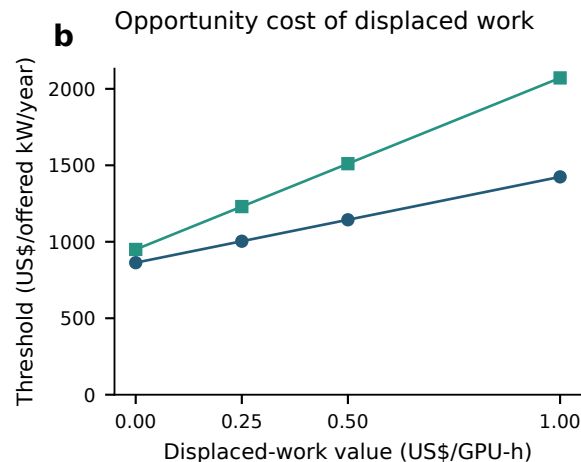
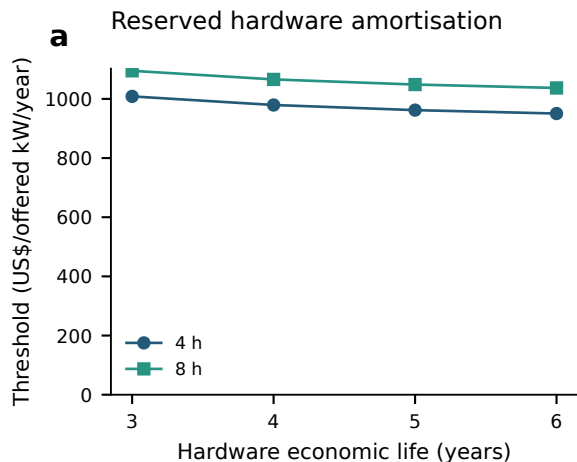
Each cell: count out of 300 independent frozen confirmation scenarios; full 5.90-kW stress request, 8-h events.
10% eligible work. Deadline multipliers apply to the reference slack. All no-response baselines pass service checks.
Failure categories overlap. Changing start spacing also moves earlier calls; this is not a pure interval-effect experiment.



a,b: minimum hosting over 100 scenarios per cell. c,d: arithmetic means over the same 100 paired scenarios.

Full-information PV planning with zero missed work; these are not causal-controller guarantees or confidence bounds.

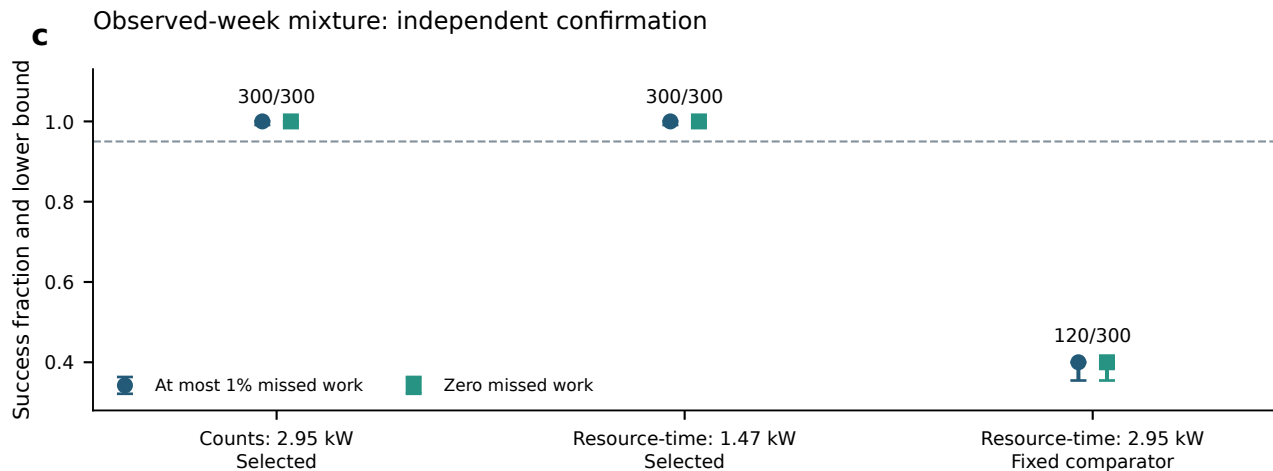
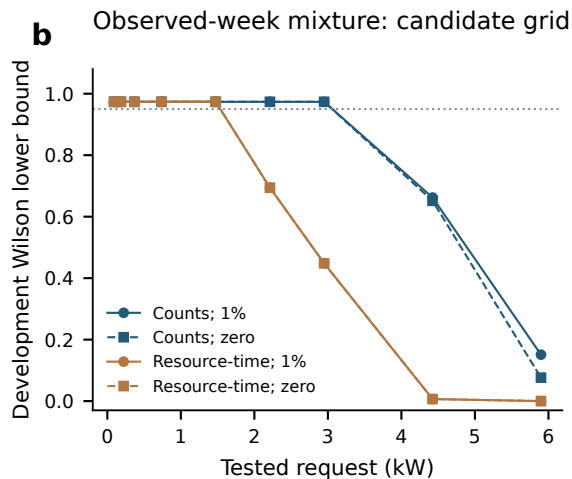
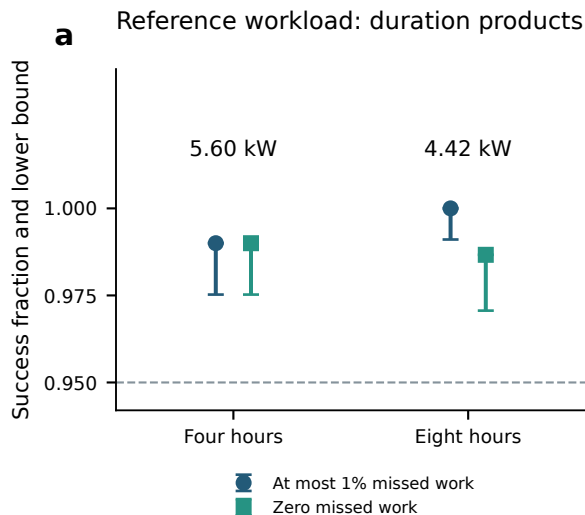
* Nearly all batch work opts in. † Business composition changes. Small storage effects may be below solver resolution.



10% eligible work; qualified single-event offers, 5.90 kW; 50 independent calls/year; proportional 1-MW accounting.

a: reserved headroom. b-d: one monetary factor at a time, without added reserve cost; other inputs at their stated reference.

Points are 95th-percentile annual-cost thresholds, not confidence bounds. These products differ from Fig. 6 repeated calls.

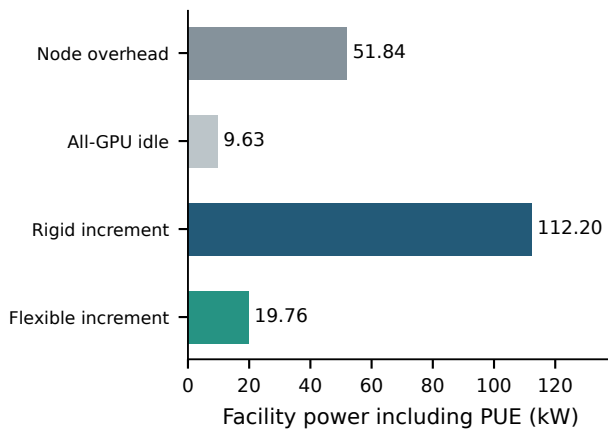


a: 300 reference-distribution scenarios per duration; retained from the earlier figure set.

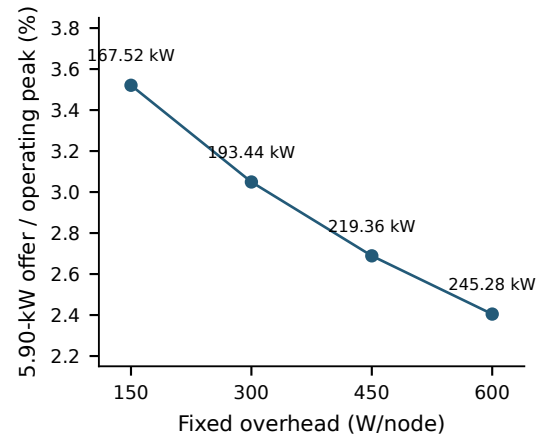
b: 100 development draws per point. c: 300 new confirmation draws per point; no confirmation reselection.

b,c condition on eight fixed weekly profiles; four eight-hour calls start 16 h apart. All bounds are pointwise.

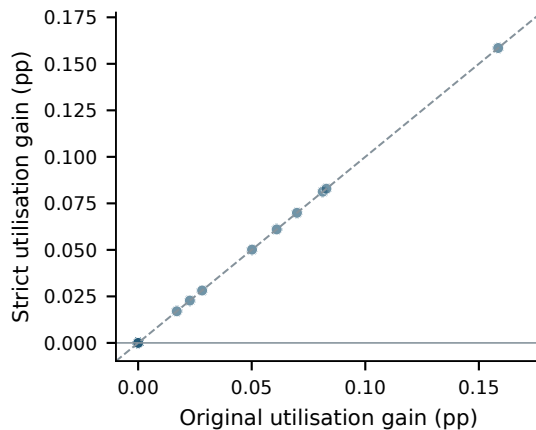
a Components of the 193.44-kW peak



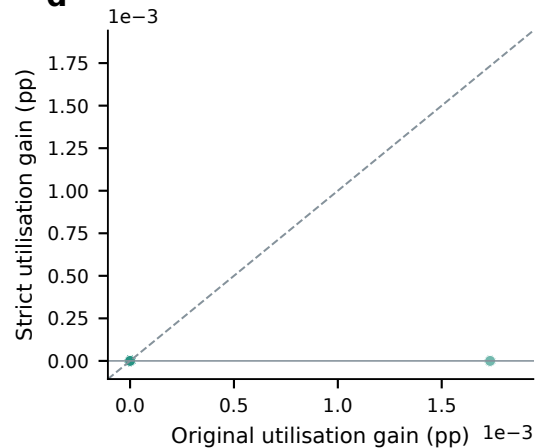
b Overhead changes the denominator



c No BESS: LP gain is unchanged



d With BESS: gain resolves to zero



a,b: deterministic accounting; all hardware and eligible work otherwise fixed. Labels in b give operating peak.
c,d: 100 paired scenarios; dashed line denotes equality. Strict relative/absolute MIP gaps $1e-7$; feasibility $1e-8$.
BESS primary PV-use bounds and the $1e-5$ -kWh lexicographic lock are distinguished in Table 24; full data retained.